

UI/UX Specialist

UCS542



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

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Assignment 5

Question1 : Write a JavaScript program to take two numbers as input from user and display their sum, difference, product, and quotient.

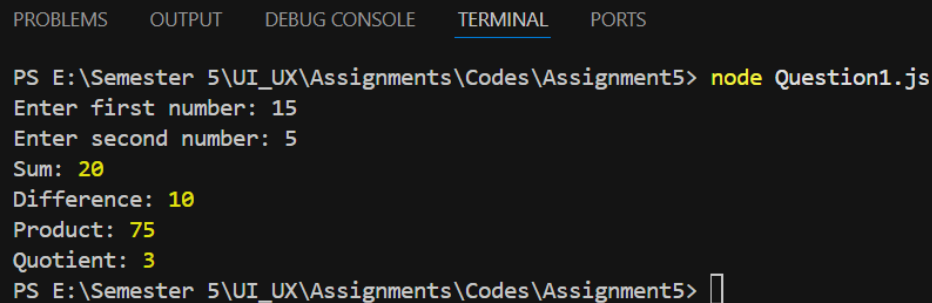
Solution:-

```
const prompt = require("prompt-sync")();

let num1 = parseFloat(prompt("Enter first number: "));
let num2 = parseFloat(prompt("Enter second number: "));

console.log("Sum:", num1 + num2);
console.log("Difference:", num1 - num2);
console.log("Product:", num1 * num2);
console.log("Quotient:", num1 / num2);
```

Terminal Output:-



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question1.js
Enter first number: 15
Enter second number: 5
Sum: 20
Difference: 10
Product: 75
Quotient: 3
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> █
```

Question2: Write a JavaScript program to create an array of 5 numbers and:

- Find the largest and smallest number.
- Sort the array in ascending and descending order.

Solution:-

```
let arr = [12, 5, 8, 20, 3];

console.log("Array:", arr);

let largest = Math.max(...arr);
let smallest = Math.min(...arr);
```

```

console.log("Largest:", largest);
console.log("Smallest:", smallest);

let ascending = [...arr].sort((a, b) => a - b);
let descending = [...arr].sort((a, b) => b - a);

console.log("Ascending:", ascending);
console.log("Descending:", descending);

```

Terminal Output:-

```

PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question2.js
Array: [ 12, 5, 8, 20, 3 ]
Largest: 20
Smallest: 3
Ascending: [ 3, 5, 8, 12, 20 ]
Descending: [ 20, 12, 8, 5, 3 ]
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> 

```

Question3 : Create a simple form with fields for name, email, and age. Write JavaScript to validate that:

- The name field is not empty.
- The email is in the correct format.
- The age is a number between 18 and 100.

Solution:-

```

<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
</head>
<body>
  <form id="myForm">
    Name: <input type="text" id="name"><br><br>
    Email: <input type="text" id="email"><br><br>
    Age: <input type="number" id="age"><br><br>
    <button type="submit">Submit</button>
  </form>
  <script>
    document.getElementById("myForm").addEventListener("submit",

```

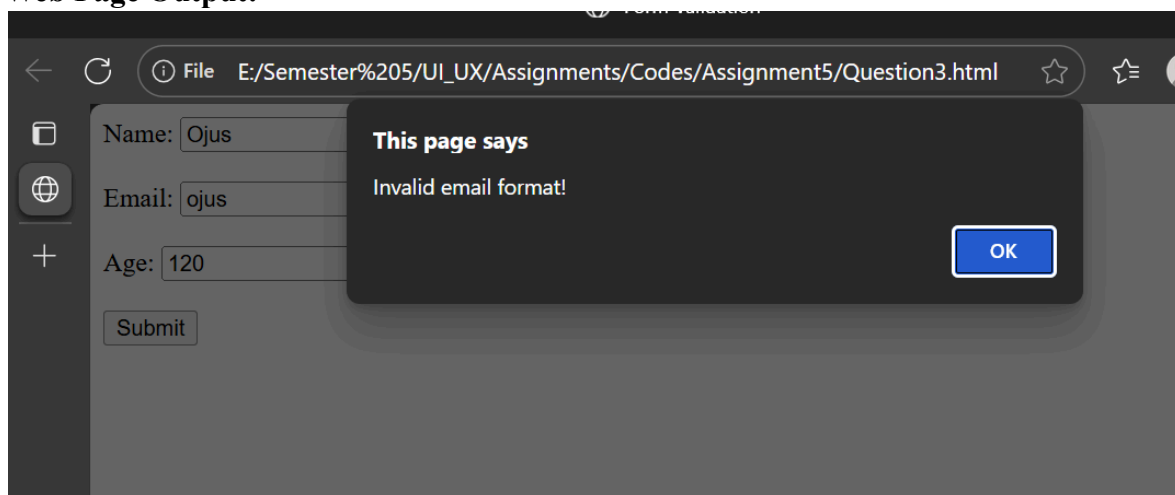
```

function(e) {
    e.preventDefault();
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let age = parseInt(document.getElementById("age").value);

    if (name === "") {
        alert("Name cannot be empty!");
        return;
    }
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    if (!email.match(emailPattern)) {
        alert("Invalid email format!");
        return;
    }
    if (isNaN(age) || age < 18 || age > 100) {
        alert("Age must be between 18 and 100!");
        return;
    }
    alert("Form submitted successfully!");
});
</script>
</body>
</html>

```

Web Page Output:-



Question4 : Write a JavaScript program to create an object representing a student (with properties like name, age, grades) and perform the following operations:

- Add a new property class.
- Update the student's grade.
- Display all the student's information.

Solution:-

```
let student = {
  name: "Ojus",
  age: 20,
  grades: "A"
};

student.class = "12th Grade";
student.grades = "A+";
for (let key in student) {
  console.log(`${key}: ${student[key]}`);
}
```

Terminal Output:-

```
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question4.js
name: Ojus
age: 20
grades: A+
class: 12th Grade
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> █
```

Question5 : Create a function that takes an array of numbers and uses the array's built-in methods (like map(), reduce(), and filter()) to:

- Remove all odd numbers.
- Multiply the remaining numbers by 2.
- Find the sum of the resulting numbers.

Solution:-

```
let numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9];

let evenNumbers = numbers.filter(num => num % 2 === 0);
console.log("Even numbers:", evenNumbers);
```

```
let multiplied = evenNumbers.map(num => num * 2);  
console.log("After multiplying by 2:", multiplied);  
  
let sum = multiplied.reduce((acc, num) => acc + num, 0);  
console.log("Sum of resulting numbers:", sum);
```

Terminal Output:-

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question5.js  
Even numbers: [ 2, 4, 6, 8 ]  
After multiplying by 2: [ 4, 8, 12, 16 ]  
Sum of resulting numbers: 40  
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> █
```